# Application Manual

**HVH Motors**:

HVH250 family

HVH410 family



Produced by Remy International, Inc.
Application Engineering
600 Corporation Drive
Pendleton, IN 46064,  USA
www.remyinc.com

## Table of Content


1. Introduction…… 3
    a. HVH250 motor family……..... 3
    b. HVH250 Output shaft options…… 4
    c. HVH410 motor family…… 5
    d. HVH410 Output Shaft Options…… 5
    e. Motor configurations …… 6
        i. Housed motor…… 6
        ii. Cartridge motor…… 6
2. Function…….. 7
    a. General Description…….. 7
    b. Typical installation requirements…….. 7
    c. Motor selection criteria……. 7
    d. Motor Design – Physical Content…….. 7
        i. Stator…….. 8
        ii. Lamination Stack……. 8
        iii. Terminal Block……. 9
        iv. Rotor…… 9
        v. Magnets…….. 10
        vi. Output Shaft…….. 10
        vii. Bearings……. 10
        viii. High Voltage Connections……. 11
        ix. Low Voltage, Signal Connections…… 11
        x. Resolver…….. 11
        xi. Temperature Sensor…… 11
        xii. Housing……... 11
        xiii. Cartridge …… 11
        xiv. Coolant Connection……. 12
        xv. Vent / Breather…….. 12
3. Performance…… 12
    a. Stall Torque…….. 12
    b. Cogging Torque…….. 12
    c. Torque Ripple…….. 12
    d. Demagnetization Temperature…… 13
    e. Thermal Guidelines…… 13
4. Integration…….. 14

a. Typical Parameters…….. 14
b. Mechanical Integration…… 14
i. Shaft / Spline Details……. 14
c. Lubrication……. 14
d. Cooling…… 15
e. B-EMF Data…… 16
f. Thermal Considerations…… 16
g. Housing integration……. 17
i. Mounting…….. 17
ii. Cartridge Motors…… 18
iii. Sleeve Type……. 18
iv. Complete Housing…….. 19
h. High Voltage Leads……... 20
i. Temperature Sensor……. 20
j. Current De-rate…….. 20
k. Wiring and Insulation……..... 21
l. HVIL……... 24
m. EMI / RF Consideration……... 24
n. Resolver……... 24
o. General Wiring Considerations…….. 25
p. Cooling System Options…….. 26
q. Cooling System Considerations……... 27
5. Glossary of Terms……..…. 27
6. Acronyms…….. 27

# 1. INTRODUCTION

This Application Manual presents details of the design and usage philosophies that are critical to the optimal implementation of Remy's High Voltage Hairpin (HVH) family motor/generators. HVH family motors can be configured to meet varied application requirements.
Contact **Remy Applications Engineering** for support in selecting the correct motor configuration for your application.

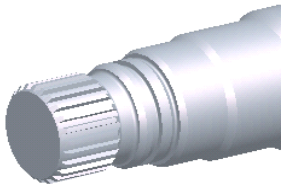
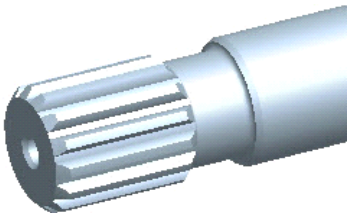
Typical markets for the HVH family of motor/generators:

- Hybrid Automotive (HEV) traction motor/generator/starter
- Fully electric vehicle (EV) traction motor
- Medium and Heavy duty automotive traction, power assist, and power generation
- Integrated Starter Generator
- On-vehicle electrical power generation
- On-vehicle, autonomous (IC engine off), mechanical power generation for accessories
- Commercial drives and generators
- Industrial drives
- Wind and hydro-electric power generation

**The HVH250 family** includes currently eight (8) configurations, determined by housing configuration, stack length, winding configuration and cooling medium

| Name | Stator | Winding | Cooling | Type |
|---|---|---|---|---|
| HVH250-90 SOM | 90 | Series | Oil | Motor Assembly |
| HVH250-90-DOM | 90 | Dual Path | Oil | Motor Assembly |
| HVH250-90-SWM | 90 | Series | Water | Motor Assembly |
| HVH250-90-DWM | 90 | Dual Path | Water | Motor Assembly |
| HVH250-115-SOM | 115 | Series | Oil | Motor Assembly |
| HVH250-115-DOM | 115 | Dual Path | Oil | Motor Assembly |
| HVH250-90-SOM | 90 | Series | Oil | Cartridge Motor |
| HVH250-DOM | 90 | Dual Path | Oil | Cartridge Motor |

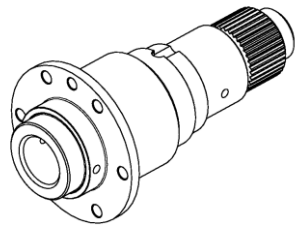
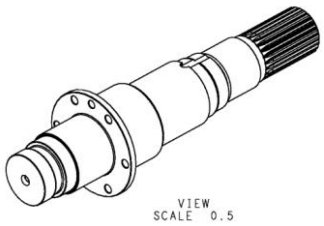
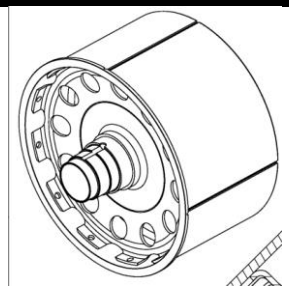
## HVH 250 Output Shaft Configurations

| VIEW | TYPE | 90 | 115 |
|---|---|---|---|
|  | **24 Tooth**<br>**Module 1.0**<br>**ANSI B92.2M** | X | X |
| | **27 Tooth**<br>**Module 0.750** | | X |
|  | **24 Tooth Internal**<br>**Module 1.0**<br>**ANSI B92.2M** | X | X |
|  | **14 Tooth**<br>**2.12 Module**<br>**ANSI B92.1-1996**<br>***(Both Ends)*** | X | |

**The HVH410 family** includes currently six (6) configurations, determined by stack length, winding configuration and cooling medium

| Name | Stator | Winding | Cooling | Type |
|---|---|---|---|---|
| HVH410-75-DOM | 75 | Dual Path | Oil | Motor Assembly |
| HVH410-75-SOM | 75 | Series | Oil | Motor Assembly |
| HVH410-75-DWM | 75 | Dual Path | WEG | Motor Assembly |
| HVH410-75-SWM | 75 | Series | WEG | Motor Assembly |
| HVH410-150-DOM | 150 | Dual Path | Oil | Motor Assembly |
| HVH410-150-SOM | 150 | Series | Oil | Motor Assembly |

## HVH 410 Output Shaft Configurations

| View: | Type: | 75 | 150 |
|---|---|---|---|
|  | *Shaft* | *X* | |
| | 38 Tooth | | |
| | External | | |
| | | | |
|   | | X | X |
| | 24 Tooth | | |
| | External | | |
| | | | |
|  | Hub | | X |
| | 38 Tooth | | |
| | Internal | | |

**HVH configurations:**

**1.Housed motor**: Designed to easily integrate into user's applications, either separately mounted or attached to a transmission or engine via standard SAE 6 mounting features.



**HVH250 motor assembly**



**HVH410 Motor assembly**

**Cartridge motor** (HVH250 only): Fully functional motor cartridge to be mounted into customer housing. Cooling oil must be provided through customer housing.



HVH 250 cartridge

## 2. FUNCTION:

### Description

The HVH family machines are internal permanent magnet, 3 phase synchronous motors utilizing distributed stator pole windings for smoother operation; and featuring rectangular cross-section hairpin windings to increase copper fill (to yield greater electrical, mechanical, and thermal efficiency). The HVH design provides a wide high efficiency range and a broad power band.

Remy HVH electric machines are suited for full four quadrant operation.

**The primary purpose of the Remy HVH motor assembly** is to convert electric energy into mechanical energy (motoring) or mechanical energy into electrical energy (generating).
Motor functionalities in vehicular applications include, but are not limited to, electrical propulsion by use of battery energy; electrical boost for an IC engine; IC engine cranking; regenerating electrical energy by electrical braking; electrical power generation from the IC engine; and conversion of electrical energy to operate a wide variety of machinery or accessories.

**Typical installations require**:

- A 3-phase motor inverter, calibrated for use in the specific motor.
- Controls and software systems capable of commanding the inverter modes of operation, inputs and outputs based on operator input and system requirements.
- External cooling system to provide either oil or WEG at appropriate flow and temperature levels

**Motor Selection Criteria:**

- Required mechanical power, and torque
- Duty cycles
- Electrical current, voltage parameters
- Operating speed range
- Coolant availability and system capability
- Mechanical interface and packaging, shaft design and mounting considerations
- Electrical interface considerations

.

### Motor Design – Physical Content

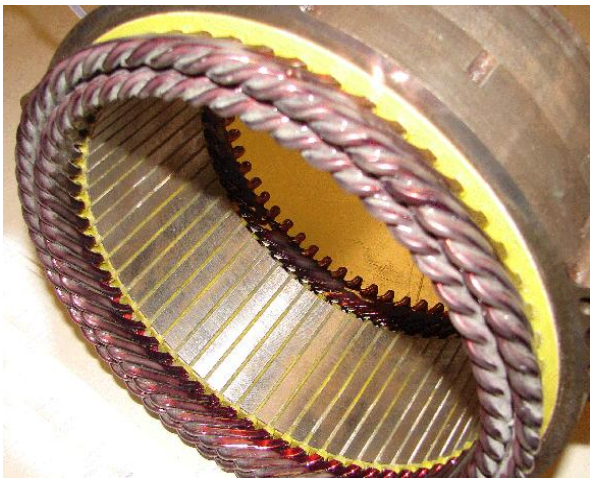
HVH family motors are made up of custom engineered components to ensure the highest performing motor in the most compact packaging for the best cost. The selection of the design parameters for the HVH Series of motors allows the motor to achieve extremely high efficiencies and torques while keeping costs in line for mass production utilization. This section details each of the primary components of the HVH

family motors as they affect the engineering process of integrating the motor into various applications

.

## Stator

Remy HVH series motor stators utilize a “High Voltage Hairpin” (HVH) winding methodology that allows high currents while operating at voltages up to 700 V (plus). The HVH design is extremely robust, providing excellent electrical and magnetic performance while remaining lightweight and compact, yielding excellent power density and thermal performance. The stator is composed of a steel lamination stack, copper conductors, and the insulation that separates the conductors from each other and the lamination steel.

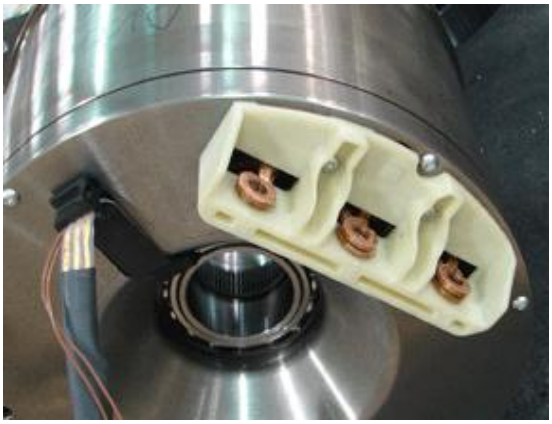


## Stator Lamination Stack

The stack of laminations comprises a rigid component that is attached to the vehicle; it maintains position as the rotor turns. The stator lamination that forms the foundation of the stator is composed of high grade magnetic steel, which optimizes the magnetic field properties of the stator, reduces eddy current losses and enhances flux control. Laminations of steel provide greatly reduced losses over a sold steel stator by limiting the magnetic flux flow perpendicular to the direction desired. (Eddy Current losses)

**Terminal Block**

Terminal block provides means to connect winding ends via a screw terminal connection to required HV cables.



High Voltage 3-Phase Connections

**Caution**:
DO NOT REPOSITION OR BEND THE HAIRPINS OR TERMINAL ENDS.

The 3-phase connections are extensions of the copper hairpins, manufactured so that each phase connection exits near the other phases and provides an appropriate termination for connection to the terminal block or customer terminals. Mechanical loading can fail the connectors.

**Rotor**:

The rotor produces torque and transfers it to the shaft.
Remy utilizes a 10 pole (5 pole pair) internal permanent magnet rotor in the HVH 250 family of motors, and a 20-pole (10 pole pair) design in the 410 family. This configuration provides excellent torque capability, low cogging effects and high efficiency while



maintaining reasonable costs. In addition, the internal (sub surface) permanent magnet configuration allows higher motor speeds without complex and expensive magnet constraint techniques.
The rotor has gone through finite element analysis and has been tested to maximum over-speed, ensuring its structural integrity well above the maximum operating speed of the motor.



**Permanent Magnets**

Remy uses a high grade rare earth magnet that provides excellent performance while being resistant to demagnetization at approved currents and operating temperatures. They are configured to provide optimal motor performance and efficiency over a wide RPM range.

**Output shaft**

Several different output shaft configurations are available. These are:

- Male spline; Standard and Metric
- Female spline; Standard and Metric

See page 4 &5 for available options

For custom shafts contact Application Engineering



**Bearings**:

The rotor turns in bearings capable of supporting the rotor mass and gyroscopic forces applied to and generated by the rotor. Radial and axial loads have to be avoided. Contact Application Engineering if you plan to load output shaft in either axial or radial direction.

**CAUTION:** The HVH motors are designed for horizontal shaft orientation; if an application requires a different orientation, consult Remy engineering.

**High Voltage Connections**
High voltage connections are made inside the HV box by means of ring terminal for each of the three phases. Ultimate wire sizes are determined by loads and safety factors, as well as distances from the motor to the power source or destination. The high voltage strain reliefs and connections are designed for 6AWG to 4/0AWG wiring. Use Industry Standards for cable connector sizing.

**Low Voltage/ Signal connections**
LV connector provides output signals from resolver and internal temperature sensor.
.

**Resolver:**
The resolver is integral part of the motor and provides precise position information to the inverter, enabling the inverter to determine the exact position of the rotor relative to the stator. Resolver signals are provided via the low voltage connection.

**Temperature sensor:**

The HVH250 motor series features a Thermistor while the HVH410 motor family uses a RTD to report critical winding temperature. Specific sensor data can be found on sheet 2 of the applicable Outline drawing.

**Housing**
The Remy designed housing is designed to provide cooling, mechanical protection, wiring support (both low and high voltage) as well as structural support to the motor.

**Cartridge option (HVH250 only)**:The various motor components can be enclosed in a cartridge that ensures alignment of the bearings, rotor, stator and resolver. The cartridge keeps the magnetic air gap between the rotor and stator – a critical design parameter – within tolerances under all operating conditions.



The cartridge motor option is designed for users who package the HVH250 motor inside a structural housing such as a transmission. Remy can be contracted to aid in the design of a suitable housing.

**Coolant connections**:
The housing has two coolant connections – one inlet and one outlet / drain for either WEG or oil cooled motors. WEG cooled motors provide cooling solely by flowing WEG around the periphery of the stator.
Oil cooled motors have cooling passages around the periphery of the stator and also have spray cooling on the end turns of the stator windings to extract heat from the windings.

**Vent:**
The housing is vented to the atmosphere through a vent on the housing.

## 3. PERFORMANCE

Performance parameters and the magnetic and thermal properties of the HVH family of motors are modeled at Remy Inc. across a range of operation to ensure that the design fundamentals are correct. These models are then validated in Remy's product development laboratory and are continuously validated on test benches of several inverter suppliers, customers, and independent laboratories.

The primary limitation in HVH series motors – as in most electric motors - is temperature. If better cooling is provided to the motor, more performance and longer life can be obtained from the motor. Even the highest grade insulation system will degrade over time with temperature a major contributing factor.

**Stall Torque**:
Remy HVH motors are capable of producing virtually their maximum published peak or steady state torque from 0 rpm to the configuration's corner speed. Note that heat generated per unit of torque can change throughout the rpm range.

**Cogging ("Detent" or "No-current") Torque**
In any permanent magnet motor, a certain amount of "no-current" torque naturally results from the magnets' poles and the stators' poles passing relative to each other. Remy has reduced this torque to a practical limit through its design and precision manufacture.

**Torque Ripple**
The motor assembly is designed to minimize torque pulsation of any mechanical or electrical frequency and their harmonics, in order to avoid exciting driveline resonance and DC voltage / current ripple. This includes magnetic cogging torque, which is reduced by the HVH series high pole count design.

### Demagnetization Temperature

Demagnetization is caused by excessive temperature and/ or excessive current. The current threshold decreases with higher temperatures.

The permanent magnets in the HVH series motors are chosen for their high magnetic flux density and Coercivity and are further protected from demagnetization by the motors' basic design. Temperature sensors are integrated into Remy motors, which signal the inverter to reduce current when pre-set thermal limits are approached.

## THERMAL GUIDELINES

Cooling information presented in the Appendix are based on Remy and customer experience. Cooling systems vary greatly, as do duty cycles and consequent heat production. Though the insulation is designed for temperatures in excess of 180ºC, the cooling system should keep the stator end-turn temperature below 160ºC (limit 180ºC), to ensure that the rotor temperature does not approach the demagnetization threshold.

# 4. INTEGRATION

## HVH MOTOR TYPICAL PARAMETERS

HVH family motors' flexible design and the wide Voltage and Current ranges paired with a broad high efficiency range make our products particular suited for use in a myriad of applications. Optimal matching of the motor to its specific application requires a study of the performance expectations, application details, duty cycle, voltage and current available, inverter selection, mounting, gearing, durability and reliability expectations, cooling capability and a wide variety of other parameters.

## MOTOR MECHANICAL INTEGRATION:

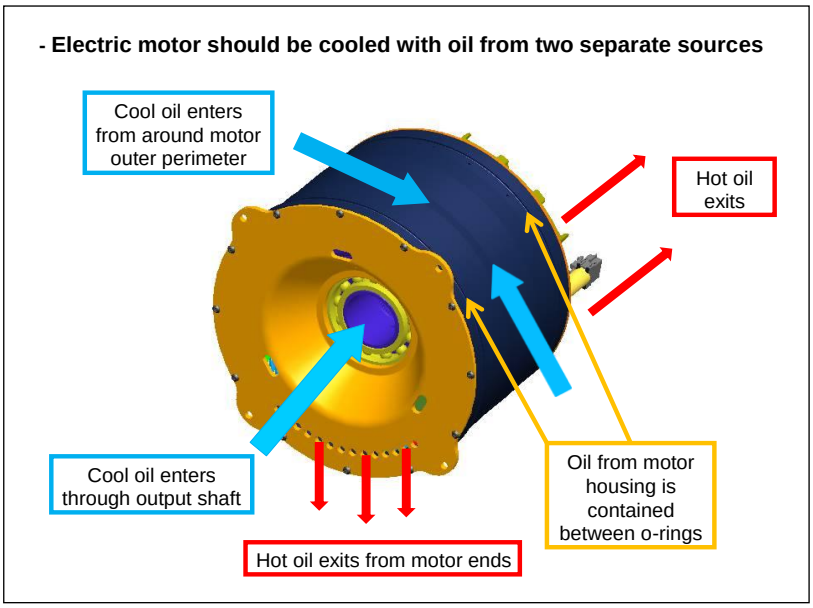
Integrating the Remy HVH family requires sufficient supporting structure to mount the motor, proper electrical connections, cooling connections and other considerations to ensure that the motor will function as expected. Cooling requirements, electrical and electronic integration, are covered later in this Section.

### Shaft / Spline Details:

Both single-end and double-end-drive shaft designs are available, as are several spline and key options.

### Lubrication:

In oil-cooled motors, bearings are oil lubricated. A filter with at least 60 micron filter rating is required for all oil cooled motors. Use only approved oils to avoid insulation break down and other negative effects. Contact Application Engineering for latest list of approved oil.




HVH250 cartridge show for demonstration only

WEG cooled motors have greased sealed bearings that require no additional lubrication or maintenance for the life of the motor.

**Cooling**:
Cooling system requirements will change depending on the application, duty cycle, and operating environment. Main variables are Cooling media – Oil VS WEG – , cooling flow and temperature.
Oil cooled HVH motors have oil-cooled stators and rotors. In oil cooled versions of the motor, oil is fed into the cooling jacket and flows from there over the end turns in the sump. This flow path ensures additional cooling to both the rotor and stator allowing higher power outputs for longer durations than would be allowable without the rotor cooling path.
WEG cooled motors provide only external cooling to the stator and leave end turns and rotor without direct cooling path. End turn Potting can be applied to improve end turn cooling.
Standard 50/50 Ethylene Glycol/ H2O solution is advised.

However, oil cooled machines typically have up to 40% better continuous performance than a WEG cooled system..

**CAUTION for oil-cooled motors only:** Ensure that oil does not collect in the air gap of the motor, i.e. between the rotor and stator. Sufficient drainage (and scavenging, in a dry sump system) during operation in any normal operating attitude must be provided to ensure that oil does not collect in the air gap. Oil accumulation in the air gap will result in demagnetization of the rotor due to extreme heat development.

**Caution**: Using oil from the IC engine's oil system to lubricate the HCH motor is not permitted, only approved oils may be used.

**Caution:** The stator end turn (measurable) limit is 160C to ensure integrity and durability of the insulation. The limit for the rotor will not be approached if the motor is operated within the guidelines provided here and by Remy engineering.

Notes: in a Remy-designed housing, cooling is provided to the motor stator via cooling passageways in the housing. In oil cooled motors, coolant flows through the cooling jacket onto the winding ends to provide additional cooling to enhance the power output capabilities of the motor.
The rotor and winding ends are not actively cooled in WEG cooled motors.

## ELECTRO - MECHANICAL PERFORMANCE IMPLICATIONS

### Back EMF

The equivalent AC RMS value is 0.785 times the DC value. (432 VAC RMS)

| | Series | Series | Dual Path | Dual Path |
|---|---|---|---|---|
| **Temp** | **90 mm** | **115 mm** | **90 mm** | **115 mm** |
| **20 C** | 45.7 | 58.4 | 22.9 | 29.2 |
| **100C** | 40.5 | 51.8 | 20.3 | 25.9 |
| **140 C** | 37.6 | 48.0 | 18.8 | 24.0 |

**HVH250 Open Circuit Line to Line Back-EMF Constant in (Vrms / krpm)**

| | Series | Series | Dual Path | Dual Path |
|---|---|---|---|---|
| **Temp** | 189.4 | 378.8 | 94.7 | 189.4 |
| **20 C** | 178.0 | 356.0 | 89.0 | 178.0 |
| **90 C** | 162.0 | 324.0 | 81.0 | 162.0 |
| **150 C** | 148.2 | 296.4 | 74.1 | 148.2 |

**HVH410 Open Circuit Line to Line Back-EMF Constant in (Vrms / krpm)**

### Thermal Considerations

Remy differentiates and publishes performance data for two different operating conditions. “Continuous” is defined as operating condition under which the heat rejection capability of the cooling system equals the heat generated by the motor. The performance under these conditions is greatly dependent on provided coolant flow and temperature.

“Peak” is defined as operating condition during which the motor temperature rises from 70 C to 180C in the course of one (1) minute. Coolant flow and temperature has very little influence on peak performance as most of the generated heat will flow into the surrounding material and greatly exceed the heat dissipation capability of the cooling system.

## REMY HOUSING MOTOR INTEGRATION

Motors purchased in a Remy supplied housing provide all required interfaces for integration.

These are:

- Coolant Ports
- LV connector
- HV conections
- Flange mounting provision
- Output shaft configuration as specified



HVH410, left; HVH250, right

**Mechanical Mounting**

The HVH250 housing mounts on an SAE 6 bolt circle, using eight M10x1.5 bolts;
the HVH410 housing uses an SAE 2 bolt circle, with twelve M10x1.5 bolts.
See Outline Drawings for more details

The mounting for the cartridge, sleeve, and / or motor housing should take into account similar factors including:

- mass of the motor, housing and attached accessories
- rotational torque provided by the motor
- maximum anticipated inertial loads (acceleration / deceleration)
- vibration / G-loads that the unit is subjected to
- gyroscopic loads induced from rotation around axes orthogonal to the motor axis
- physical location of the motor in relation to the chassis and other drive components
- sufficient safety factors

**Cartridge Motors (HVH250 family)**

For motors purchased as a cartridge unit, a customer designed housing must be provided. The cartridge is a machined steel enclosure that provides sufficient structure for holding the stator, rotor, bearings and resolver in alignment. It also contains all features needed to provide adequate cooling assuming sufficient oil flow around the outer perimeter. – customer's responsibility

The cartridge motor option accommodates users to package active power components of the motor into a custom housing.. The customer is then responsible to ensure that the requirements of mechanical mounting, electrical insulation and wiring, cooling and lubrication are met. Remy Engineering can assist with proper design practices and recommendations.

The cartridge housing is designed for mounting in a cavity with a sealed oil bath circulating around the circumference of the cartridge. This oil bath is critical, providing sufficient cooling of the motor. The o-ring mating surface for the cartridge must have a surface finish of Ra: 3.2 or better to ensure sufficient seal quality and prevent excessive leakage or o-ring damage during installation and operation.

Please refer to the model and drawings provided by Remy Applications Engineering of the cartridge housing for additional details.

**Sleeve Type Mounting (HVH410 only)**

Sleeve type mounting configurations are similar to the cartridge type assembly but do not employ the use of covers and bearings. The sleeve facilitates ease of assembly but requires that users properly design their bearing system, resolver mounting system, and sleeve retention system. The sleeve type mounting system is not available on HVH250 motors, but is available on HVH410 motors.

Note: Remy Engineering can evaluate customer applications on a case by case basis to ensure that the Remy motor is properly applied in the customer vehicle.
Remy Engineering evaluation and operation of thermo-coupled motors in the intended customer application and operating environment is necessary for Remy to project actual motor life and grant the customer a motor warranty.

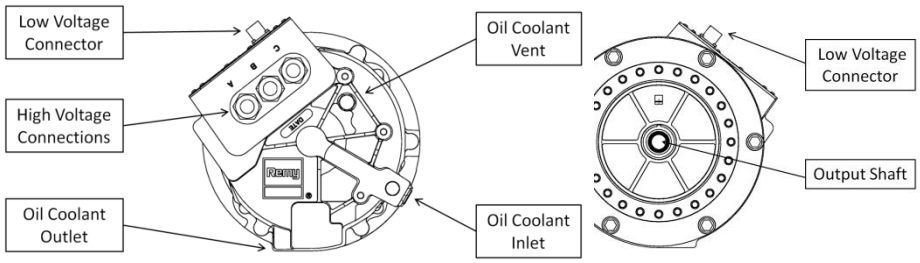
## Remy Housing

The HVH250 family is available in a Remy designed housing that can ensure proper alignment, protection, cooling, and lubrication.

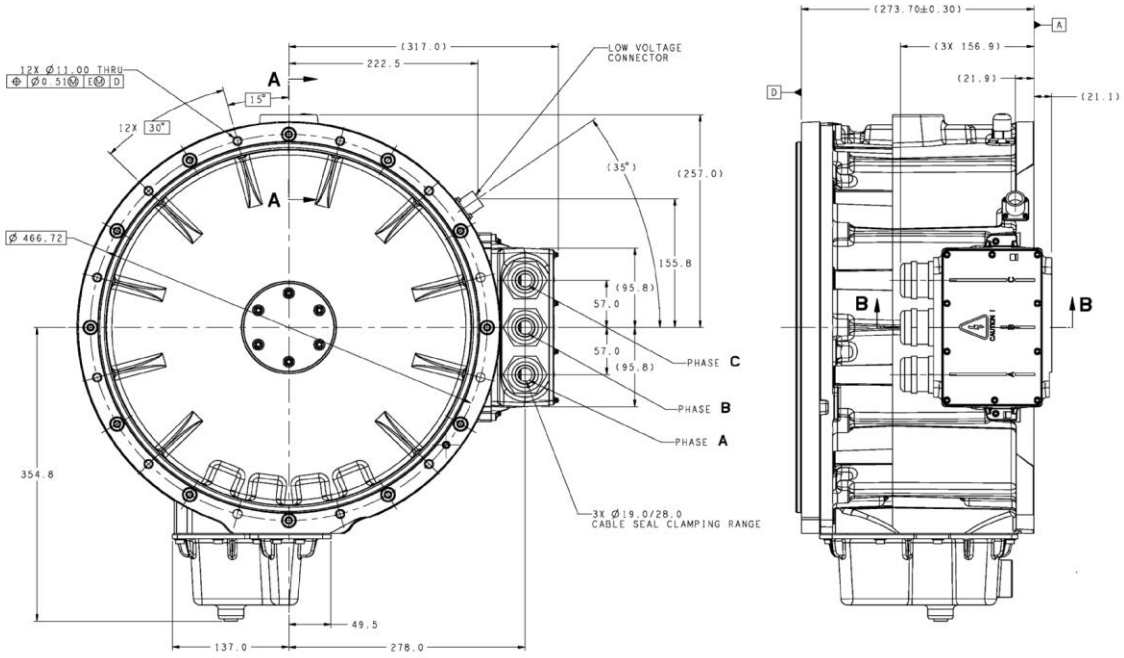



emy housing is designed for utility and
tion, and meets all design
ements for mechanical and electrical
ctions, lubrication, and environmental
tions.

oil inlet/exit
ric oil-cooled housing shown; HVH250
')




The HVH410 family (below) is also available in a Remy housing.




No cartridge option is available for the HVH410 family.

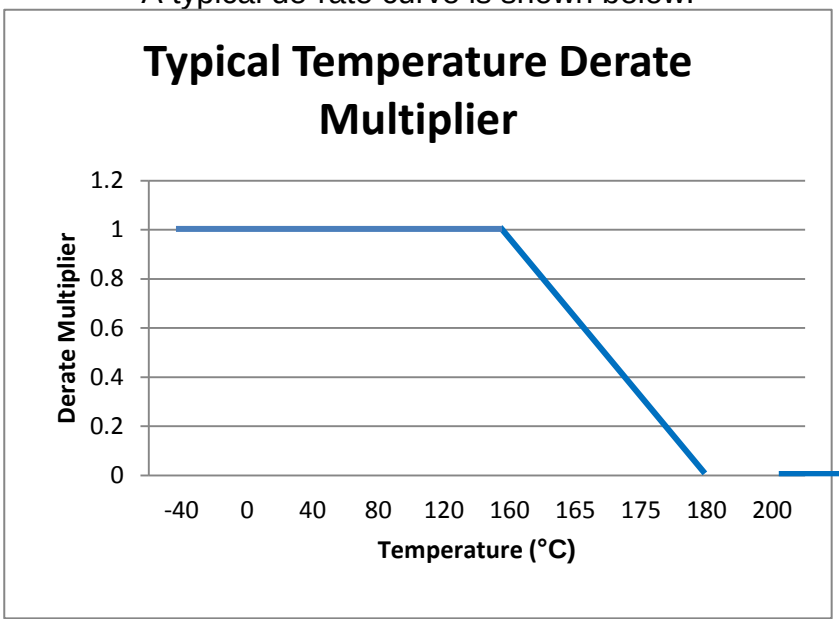
**High Voltage Leads**:
The high voltage leads (phase leads) carry up to 600 amps and 750 volts in a modulated sine wave. Normal inverter modulation frequencies vary from 4 kHz to12kHz. Exact values are dependent on the inverter used for controlling the motor and its calibration.

**WARNING: The customer is responsible for ensuring that all high voltage wiring is adequately marked, insulated and protected to prevent high voltage wiring from accidental contact. In addition, the inverter system providing the high voltage must be able to detect any failure and disconnect the motor and its associated wiring from the voltage source to prevent electrocution or short circuits.**

**Temperature Sensor**
All motors are equipped with either a RTD (HVH410 Series) or a Thermistor (HVH250 Series) to sense motor temperature. These temperature sensors must be incorporated into the inverter control algorithms to reduce output should excessive temperatures be produced. Typical algorithms reduce current starting at 160C and linearly de-rate to 0 Ampere at 180C.

**CURRENT DERATING**

A typical de-rate curve is shown below.




**Caution** Warranty coverage will not be provided for motors where temperature limits have been exceeded.

Note: motor-specific parameters for HVH motors, including the Ld and Lq tables at several temperature set-points, are available from Remy Applications Engineering. Knowledge of these parameters will allow accurate modeling of the electrical and power properties of the HVH family motors. A signed NDA is required for this information.

## Electrical Protection / Packaging

**DC Voltage Isolation**:
The Motor Assembly must be isolated between the low and high voltage systems.

**WARNING: This manual assumes that all personnel potentially exposed to any high voltage source be properly trained and equipped, and follow necessary safety precautions and protocols.**

**The customer is responsible to ensure that the mounting of the housing or cartridge is sufficient to ensure that no structural failure or unconstrained condition may occur. Failure of the mounting system may result in system damage or bodily harm due to the large amounts of both mechanical and electrical power becoming unconstrained.**
**Please contact Remy Applications Engineering with any design concerns.**

## Electrical Insulation and Wiring

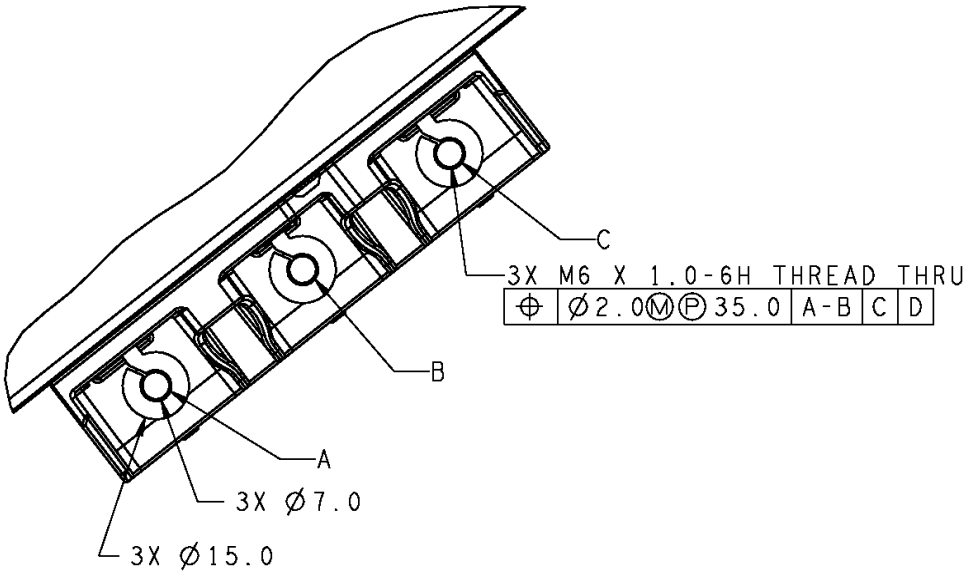
High voltage connections can be dangerous. In some cases, these wires may carry over 700VDC and 600 amps of current. Regardless of the current or voltage level, care should be taken to ensure safe operation.

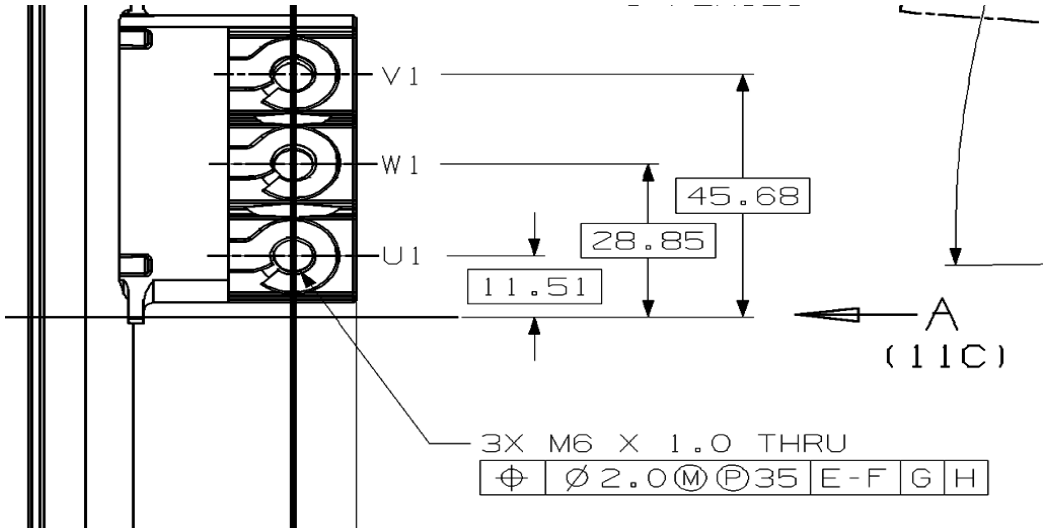
It is imperative that the terminals are protected from any risk of shorting or damage by utilizing the recommended connectors and good design engineering.

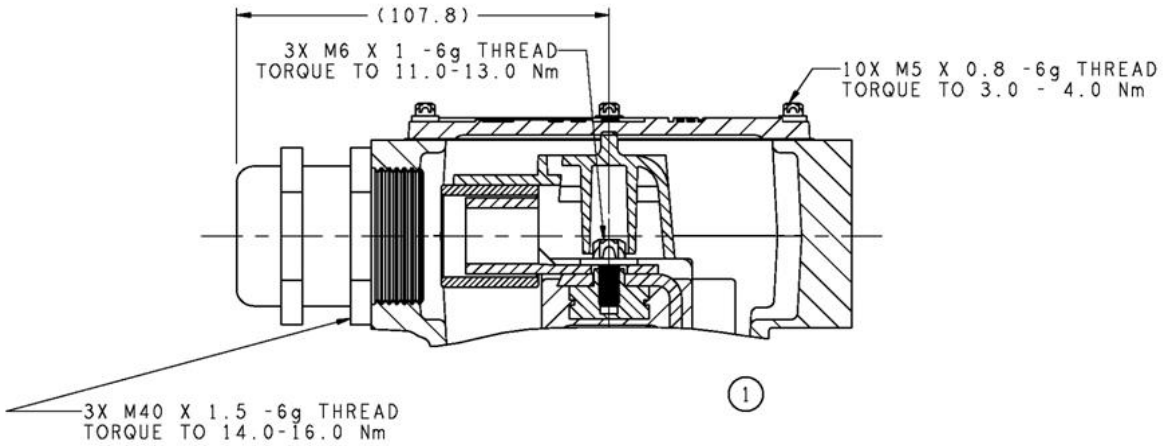
The 3 phase connections are labeled **A, B**, and **C** and are connected with **M6x1.0** bolts.

The recommended torque is **11 +/-2 Nm (~8 lb-ft).** It is also important to ensure that the wires are properly strain relieved and supported to ensure that no more than a **50 N (~ 11 lb or 5 kg**) push or pull force is ever applied in any direction.

The figures below are for illustration purposes only.

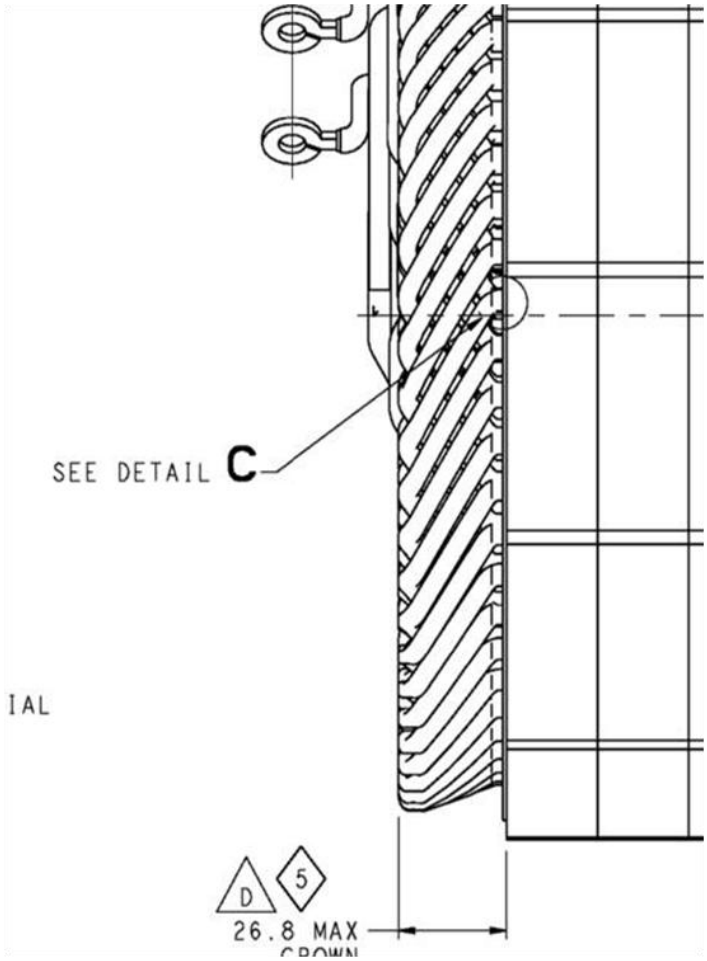



HIGH VOLTAGE CONNECTION

**Caution:** Care should be exercised to ensure that the phase connections are properly made between the inverter and motor or motor damage may result.




The **Low Voltage Connection** provides connection to resolver and temperature signals
The low voltage pin-out details are found on sheet 2 of the motor specific Outline Drawing

**WARNING: The customer is responsible for ensuring that all high voltage wiring is adequately marked, insulated and protected to prevent high voltage wiring from accidental contact.**

**In addition, the inverter system providing the high voltage must be able to detect any failure and disconnect the motor and its associated wiring from the voltage source to prevent electrocution or short circuits. Additional safety measures may also safely shutdown the vehicle engine and / or prevent restarting the engine in the event of a HV system or sensor failure.**

**High Voltage Interlock (HVIL)**

For most motor models Remy offers a switch to accommodate a HVIL system. The switch is integrated into the HV connection box and is activated by the postion of the HV box cover. When the cover is installed and preventing easy access to the HV terminals the contact is closed and the switch provides pin to pin continuity. With cover removed switch contact is opened and circuit continuity interrupted.
It is important to notice that the switch itself does not turn off any HV source, but only provides a fail-save signal to be used by the control system (inverter) to act upon!

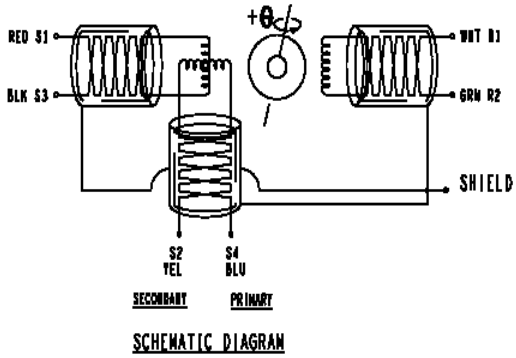
**WARNING: HVIL output signals should be routed to the appropriate inverter inputs. Do not attempt to defeat or disable the HVIL signal or its function. This is a critical safety function that shall be implemented and appropriately tested prior to motor operation.**

**EMI / RF Considerations**:

Remy HVH series motors are inherently tolerant of EMI and RF noise and only in rare instances can outside EMI and RF affect their operation. Due to the fact that HVH family motors operate at extremely high voltage and current levels, production of EMI and RF noise can and will occur unless appropriate design considerations are taken into effect. Whenever problems are encountered or suspected Remy recommends the consultation of experienced engineers with applied knowledge of solving EMI and RF based issues. The methods and techniques used to attenuate EMI and RF noise to a level that is suitable for a particular application vary greatly and are highly dependent on many factors. Careful wiring layout, proper shielding and grounding techniques, and aggressive filtering and attenuation methods may all be required to ensure that a particular system does not suffer any anomalies from EMI and RF noise.

The **Resolver** uses low voltage leads to its three coils, which determine motor position. It does this by having one phase energized with a sine wave signal at a fixed frequency and comparing the transformation ratio output in the other two coils. The other two coils (sine and cosine) are arranged such that the transformation ratio outputs are 90 degrees out of phase from each other. This enables the decoding circuitry in the inverter to determine the precise motor position from a standstill up to very high speeds.

The sine and cosine signals are prone to noise due to the nature of their being coils operated at relatively low voltages and current and utilized by sensitive signal processing circuitry. Careful design work has been completed by Remy to ensure that any crosstalk between the phase leads and the low voltage leads is minimized. This involves internal shielding of the wires as well as careful magnetic design to ensure that false signals are not created.



SCHEMATIC DIAGRAM

**General Wiring Considerations**:

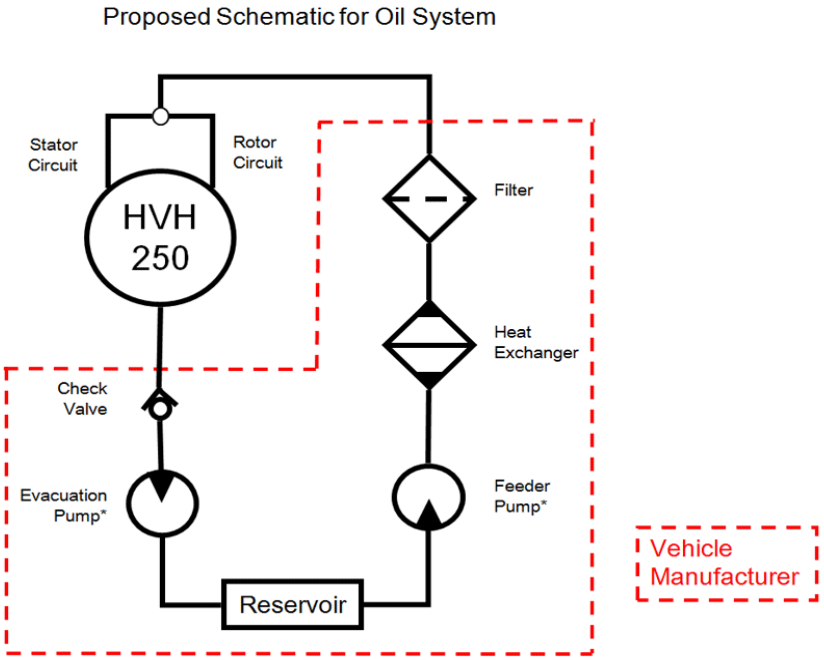
- Route the HV leads and LV leads separately. Preventing these two sets of wiring from running in close proximity will reduce noise and prevent crosstalk between the signals. Remy recommends a minimum of 6 inches between the phase (HV) leads and other signal wires.
- Route the HV leads away from other data and signal lines, as well as other low power lines that may be susceptible to noise or interference. Remy recommends a minimum of 6 inches between the phase leads and other signal wires.
- Separate the HV leads from other low voltage wiring with metal components such as a frame rail or other sheet metal, being careful to avoid placing the HV leads where they may be prone to thermal or mechanical damage.
- Always use orange cable / conduit on HV lines.
- Ensure that a shielding technique is utilized on the HV leads to reduce radiated electrical or magnetic noise. Shielded wire is highly recommended.
- Ensure that the resolver signals are properly extended. Remy utilizes shielded twisted pair for each of the 3 resolver signals and provides a pin on the connector to enable carrying the individual shields through to the customer wiring harness.
- Ensure that the Thermistor wiring is protected from both common and differential mode interference from neighboring wiring by appropriate EMI / RF mitigation techniques - shielding and / or twisted pair are valid methods to reduce noise on these lines.
- Minimize the length of all wire runs that are prone to transmission (HV phase leads) or injection (all low voltage, signal and power leads). Minimizing length reduces the available length to transmit or inject. However, do not sacrifice wire separation to reduce length.
- Ensure all associated circuitry is resistant to noise and has appropriate filtering.

# Cooling System Options

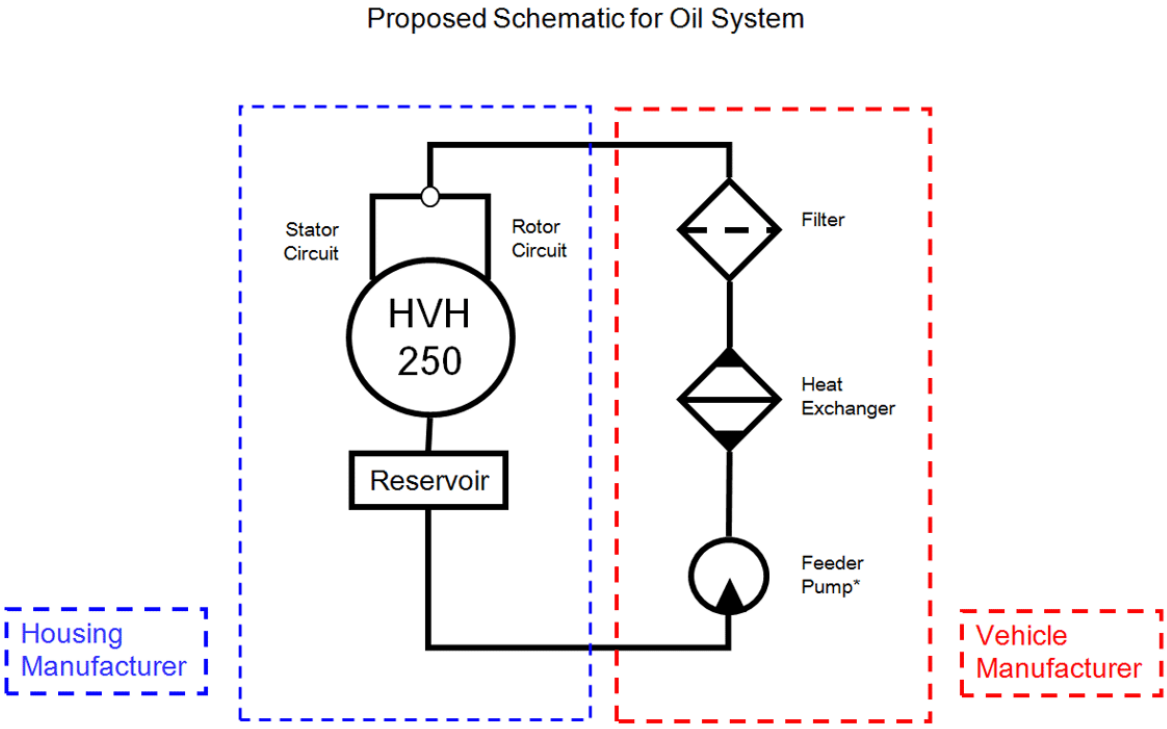
## Basic hydraulic schematic, Dry Sump Option:

**Design Criteria**

- Evacuation pump to have higher capacity than feeder pump to prevent accumulation of oil inside motor
- Oil must not flood the motor when pumps are not operating
- Oil flow has to be assured in cold conditions




## Basic hydraulic schematic, Wet Sump option:



**Wet and Dry Sump Oil System Configurations** (representative)

**When a customer designs a housing**, cooling system design and configuration considerations must include:

- fluid type,
- cooling path,
- thermostat settings,
- heat rejection capability,
- pump flow capability,
- pressure drops through the system.

It is imperative that the customer ensure that the motor is operated within approved temperature limits.

## 5. Glossary of terms

- Generating: Conversion of mechanical energy into electrical energy.
- Inverter: The power supply device that converts DC to AC. The inverter uses rotor position information provided from the resolver to accurately determine the rotor pole position with reference to the stator poles.
- Magnetic Air Gap: Critical design distance between rotor and stator
- Speeds:
  - Maximum operational speed: The maximum allowable continuous motor speed at normal operating voltages.
  - Maximum non-operational speed: The maximum motor speed that will not result in permanent deformation or damage to motor components.
  - Maximum structural speed: The maximum speed the motor may attain without incurring extensive and immediate damage.
- Motoring: Conversion of electrical energy into mechanical energy, e.g., driving the wheels of an EV.
- Resolver: Senses the relative positions of rotor and stator
- Rotor: The rotating portion of the motor
- Stator: The static, wound portion of the motor

## 6. Acronyms

- AC – alternating current
- DC – direct current
- EV – electric vehicle
- FMEA -- (Failure Mode Effects Analysis)
- HEV – hybrid electric vehicle (typically, a vehicle using both an IC engine and an electric motor/generator)
- HVH – High Voltage Hairpin
- HVIL -- High Voltage Interlock
- IC – Internal Combustion
- RTD -- Resistive Thermal Device
- WEG -- Water Ethylene Glycol (cooling medium or system)